Comprehension of Japanese passives:

An eye-tracking study with 2-3-year-olds, 6-year-olds, and adults

Miwa Isobe[*], Reiko Okabe[**], Yukino Kobayashi[***], Shigeto Kawahara[†], Tomoko Monou[††],

Kazuhiro Abe[†], Rei Masuda[†], Saeka Miyahara[†] and Yasuyo Minagawa[†]

[*] *Tokyo University of the Arts,* [**] *Senshu University,*

[***] *Tsukuba University of Technology,* [†] *Keio University,* [††] *Mejiro University*

1. Introduction

Previous acquisition studies within the framework of generative grammar have shown that, while children with different language backgrounds acquire many properties of their native grammars by around age 4, they often have difficulty with comprehending passive sentences cross-linguistically even between the ages of 5 and 6. Several experimental studies on passives in child Japanese have demonstrated that children as old as 5 or 6 exhibit non-adult-like interpretations of passives with a *ni*-phrase, equivalent to a *by*-phrase in English (e.g., Sugisaki 1999; Minai 2000; Sano et al. 2001). These studies report that children interpret passive sentences as if they were active sentences. On the other hand, it has also been reported that Japanese 4-to-6-year-olds can correctly interpret passives without the *ni*-phrase (e.g., Okabe & Sano 2002; Ishikawa et al. 2018), and that even children between the ages of 2 and 3 can produce short passives, albeit with a limited number of verbs (Harada & Furuta 1999). These previous studies raise the possibility that children as young as 2 or 3 have already acquired basic knowledge of Japanese passives but that experimental methods prevent researchers from extracting such young children's knowledge of passives. To our knowledge, no experimental studies have been attempted to explore the comprehension of Japanese passives by children around the age of 3 since few reliable methods were available to test young children's comprehension of this construction.

In light of this situation, the present study investigates whether Japanese-speaking 2-3-year-olds can distinguish short passives from their active counterparts by employing the preferential looking paradigm using an eye-tracker, which allows children's sentence comprehension to be examined in a non-intrusive manner. We compare young children's eye-gaze data with those of 6-year-olds and adults. Although the previous acquisition studies on Japanese passives have reported that 6-year-olds are perfectly adult-like, no studies have ever reported gaze data of 2-3-year-olds or of 6-year-olds and adults. Our experiment revealed that the eye-gaze behaviors of young children were different from those of 6-year-olds and adults. The fixation time data suggest that children age 2 to 3 were able to comprehend active sentences but had difficulty in dealing with passives in the same way as older children. The results of the experiment also raise the possibility that younger children can notice the passive morpheme, which is not included in an active sentence, but cannot promptly revise the initial interpretation guided by the typical interpretive strategy of taking the first NP as the agent of the verb.

2. Passives in English and Japanese

This section briefly surveys the syntactic analyses of passive constructions in both English and Japanese, on which we base our experimental study concerning the acquisition of passives.

An example of the English passive is given in (1a), and its underlying structure is shown in (1b) along with illustrations of an A-chain and theta-transmission, following Jaeggli (1986).

(1) a. The teacher was tickled by the child.

A-chain

b. The teacher [e] **was** tickl**ed** [e] **by** the child.

theta-transmission

In the modern generative grammar, the English passive sentence is considered to include promotion of the object to subject position through A-movement. Sentence (1a) indicates that the object NP of the verb *tickle* moves to the subject position on the surface structure, and the external theta-role of the verb which is absorbed by the passive morpheme *-ed* is optionally discharged to the NP in the *by*-phrase. Passive sentences with a *by*-phrase are widely called *long passives* and those without it are called *short passives*, and we will adopt these terminologies in our study.

We will now see that a similar analysis can be applied to Japanese passives as well. It is well-known that passives in Japanese are largely divided into two types: direct passives and indirect passives (e.g., Shibatani 1978). Although both types are similar in that they involve the passive morpheme *-(r)are*, it is argued that the former corresponds to the English passive, while the latter, which is formed by adding an extra argument to the active sentence, does not have an English counterpart. Since we will discuss the acquisition of Japanese passives in parallel with English passives and our experiment deals with only direct passives, we will focus only on that type of passive henceforth.

An example of a Japanese active sentence is given in (2) with its passive counterpart shown in (3).

(2) *Active*

Kodomo-ga sensei-o kusugut-ta.

child-Nom teacher-Acc tickle-Past[1]

'The child tickled the teacher.'

(3) a. *Passive*

Sensei-ga kodomo-ni kusugur-are-ta.

teacher-Nom child-by tickle-Pass-Past

'The teacher was tickled by the child.'

A-chain

b. Sensei-ga [e] kodomo-**ni** [e] kusugur-**are**-ta.

theta-transmission

The Japanese passive sentence in (3a) is posited to have the underlying structure in (3b) (e.g., Kubo 1992). Like English passives, Japanese passives are assumed to involve A-chain formation and theta-transmission. The passive suffix *–(r)are* is attached to the verb stem *kusugur-* 'tickle,' and the object of the corresponding active sentence, *sensei* 'teacher,' appears as the subject with nominative case *–ga*. The suppressed theta-role of the subject in the active sentence, *kodomo* 'child,' can optionally appear with the dative marker *–ni* 'by,' just as in the English *by*-phrase.

3. Child Passives

3.1. Acquisition of passives in English and other languages

Previous acquisition studies have shown that children exhibit non-adult-like interpretations of long passive sentences cross-linguistically until around the age of 5 or 6. In particular, several sources have been proposed for the delay in understanding English passives. Borer & Wexler (1987) propose that children's ability to form an A-chain undergoes maturation and until the ability matures children treat passives as adjectival ones. Fox & Grodzinsky (1998), on the other hand, propose that children have difficulty in discharging the external theta role to the *by*-phrase, not in forming an A-chain. Their claim is based on the results of their experiment, which revealed that children performed well on passives with actional verbs such as *chase* and *touch*, and they made mistakes only on long passives with non-actional verbs such as 'The boy is seen by the horse.' Although both Borer and Wexler (1987) and Fox and Grodzinsky (1998) attribute children's difficulty with passive comprehension to their non-adult-like grammatical knowledge, some recent studies argue against the grammatical accounts for this difficulty. O'Brien et al. (2006), for example, showed experimentally that comprehension of English passives by children as young as ages 3-4 was adult-like when the construction was used in a pragmatically felicitous context, suggesting that the children had already acquired adult-like knowledge of passives. Recently, an incremental processing account for children's difficulty in comprehending English passives has been proposed by Deen et al. (2018), who claim that children's processing mechanisms are still in the process of developing while they may already have developed adult-like knowledge of passives.

Moreover, many acquisition studies on passives in a wide variety of languages other than English have reported that passives are difficult for children cross-linguistically (e.g., Spanish (Pierce 1992), Greek (Terzi & Wexler 2002), and Russian (Babyonyshev & Brun 2004)), although there is evidence that children acquiring Sesotho have adult-like knowledge of passives by around the age of 3 (Demuth et al. 2010). Recently, Armon-Lotem et al. (2016) conducted experiments with 5-year-olds across different European languages and found that they performed well on short passives but showed difficulty with long passives in languages such as Catalan and Hebrew.

3.2. Acquisition of passives in Japanese

It has also been reported that Japanese-speaking children have difficulty with passives and tend to interpret passive sentences like (3a) 'The teacher was tickled by the child' as if they were active, i.e., 'The teacher tickled the child' (e.g., Sugisaki 1999; Minai 2000; Sano et al. 2001; Okabe & Sano 2002; Sano 2013). Some experimental studies, such as Sugisaki (1999) and Minai (2000), claim that children's difficulty in understanding Japanese passives is caused by their immature grammatical ability to form an A-chain following Borer & Wexler (1987). On the other hand, studies such as Sano et al. (2001) and Okabe & Sano (2002) have shown experimentally that Japanese-speaking children around the age of 4 can comprehend short passives but have trouble interpreting long passives. This suggests that children already have the ability to form an A-chain: Otherwise they could never interpret short passives which involve the movement of an object NP to the subject position, i.e. an A-chain. These studies claim that children already have the ability to comprehend passives but have difficulty with interpreting the agent role in *ni*-phrases, i.e., '*by*-phrases.' Their experimental results are compatible with Fox & Grodzinsky's (1998) claim. Recently, Ishikawa et al. (2018) reported a similar observation: if agent *ni*-phrases were omitted, the comprehension of passives by 4-year-old and older children improved compared to comprehension of long passives with an agent role. In addition, analyses of spontaneous speech corpora have reported that utterances by children at age 2 contain short passive sentences, although they are limited in the variety of verbs used in their speech (Harada & Furuta 1999).

Considering the findings from these previous studies, it might be possible that children younger than 4 years of age can comprehend short passives. Yet, comprehension of Japanese passives by children as young as ages 2-3 has rarely been experimentally tested, partly because the tasks adopted for experiments on child passives are generally not suitable for children younger than 4 years old. The previous experimental studies on the acquisition of Japanese passives adopted methods such as the act-out task (e.g., Sano 1977; Hakuta 1982; Murasugi & Kawamura 2005), the picture-selection task (e.g., Sugisaki 1999; Minai 2000), or the truth-value judgment task (e.g., Sano et al. 2001; Okabe & Sano 2002). These tasks place an extra cognitive burden on young children, which may hinder a precise understanding of their comprehension of passives (see Goodluck 1996: 152 for an example of the discussion on the cognitive complexity of the act-out task). The fact that they can produce some passive sentences in their spontaneous speech raises the possibility that they may be able to understand passive sentences but that technological limitations may have prevented us from revealing their grammatical knowledge. To address this gap, we conducted an eye-tracking study on the acquisition of Japanese passives, which we believe sheds new light on the behavior of younger children.

4. Experimental study

4.1. Research question

Given the findings of previous acquisition studies, the research question given in (4) arises concerning young Japanese-speaking children's comprehension of passives.

(4) Research question:

Do Japanese-speaking children as young as ages 2-3 show eye-gaze behavior similar to older children and adults when presented with passive sentences along with visual stimuli depicting matching and mismatching scenes?

4.2. Participants

We tested 60 children ranging in age from 2;6 to 3;5 (mean=2;10). We also tested 25 children of age 6 (mean=6;7) and 10 adults for comparison. We selected 6-year-olds in reference to Ishikawa et al. (2018): In their experiment, the participants of age 6 correctly comprehended Japanese short passives 94.0% of the time. All of the participants were from Tokyo or surrounding areas, and all the participants of ages 2-3 and age 6 were female because this experiment was conducted as part of a larger project on cognitive development which required female participants.

4.3. Method

The task we adopted was a preferential looking task in which two short videos followed by static images of their respective final scenes were presented side-by-side to each participant. We recorded participants' gaze data using a Tobii X120 eye-tracker. Each participant was asked to sit in front of a monitor and watch the screen very carefully. The timeline of a trial with sample auditory and visual stimuli is shown in Figure 1.

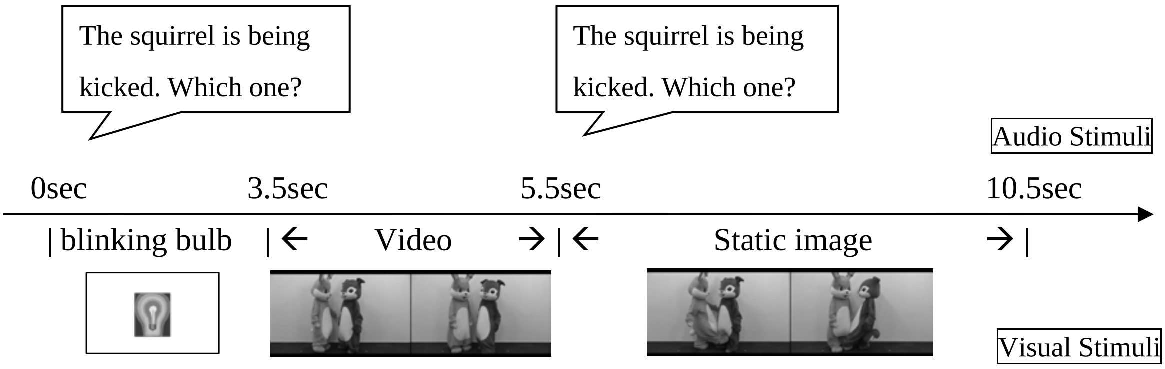

Figure 1: Timeline of a trial

Each trial lasts 10.5 seconds and begins with an alerting sound along with a blinking light bulb intended to attract the child's gaze to the center of the monitor. After the alerting sound, a test sentence, such as "the squirrel is being kicked," followed by "which one?" is played for 2.5 seconds with an animation of a light bulb. After the test sentence is auditorily presented, two animations with two animals begin playing side-by-side on the split-monitor. One of the animations matches the situation depicted by the audio stimulus, while the other mismatching animation shows a scene where the agent and the patient are reversed. The animations are played for 2 seconds, after which the test sentence is played for the second time. The animation becomes a static image showing the outcome of the action, which enables children to remember what has happened to the two animals in the scene. The static images are shown on the monitor for a total of 5 seconds.

We decided to present the stimulus sentences auditorily before any visual stimuli appeared on the monitor for the following reason: It was often the case in previous experimental studies that child participants looked at pictures or stories before the test sentences were presented and so were likely to choose a picture or judge the truth-value of a sentence even before they finished listening to the whole sentence. Note also that the passive morpheme *–(r)are* appears at the very end of the sentence following the verb stem, which makes the passive morpheme less conspicuous and easily ignored. Therefore, in order to ensure that child participant would choose a picture after encountering the passive morpheme, we decided on the ordering of auditory and visual stimuli shown in Figure 1.

Additionally, each test sentence was played twice in order to help the child review her initial interpretation, which was made just after the first presentation of the sentence. This is in line with the results of an experiment on child passives by Deen et al. (2018): Their experiment revealed that English-speaking children's comprehension of passives improved when test sentences were given twice, suggesting that repeating test sentences might enable children to review an initial interpretation while listening to the test sentence for the second time.

Before the test session, each of the children between ages 2 and 3 was shown pictures of four animals which appeared in the animations and asked to name the animals to be used in the test sentences. If a child was unable to give the correct names of the four animals, an experimenter told her the names and asked her to repeat them, and then asked her to say the names of the animals several times while showing her each picture, like a flash card game. The child then went through a pretest in which she familiarized herself with the procedure and all the transitive verbs used in the experiment, i.e. *kick* and *push* for test sentences and *hug* and *pat* for fillers. The pretest sentences consist of an overt verb in a progressive form with a null object, such as “the squirrel is kicking (*pro*),” followed by “which one?” Each trial in the pretest session lasts for 7.5 seconds. After the alerting sound, the pretest sentence is played for 1.5 seconds with an animation of a light bulb. Then two animations with two animals begin playing side-by-side on the split-monitor. One of the animations matches the situation depicted by the audio stimulus, while the other mismatching animation shows a scene where the agent’s action toward the patient is completely different from the target action. The animations are played for 2 seconds before becoming a static image showing the outcome of the action. The static images are shown on the monitor for 3 seconds.

4.4. Test Sentences

We tested a total of 16 sentences: 8 target sentences, including both active and passive sentences, along with 8 filler sentences.[2] Among the 8 test items, 4 contained the verb *keru* ‘kick’ and the other 4 contained the verb *osu* ‘push.’ Examples of both active and passive test sentences are given in (5) and (6) respectively. Note that active test sentences were given without overt direct object NPs, which is entirely acceptable in Japanese, a null argument language. Passive sentences were all short passives, i.e., passives without *-ni* ‘by’ phrases. This allowed both active and passive test sentences to consist of the same number of words, making them minimal pairs that differed only in the presence/absence of the passive morpheme.

(5) *Test sentence with verb* keru *‘kick’*:

a. Risusan-ga ket-teru yo. Dotti? (Active)

squirrel-Nom kick-ing excl which

‘The squirrel is kicking *pro*. Which one?’

b. Risusan-ga ke-rare-teru yo. Dotti? (Passive)

squirrel-Nom kick-Pass-ing excl which

‘The squirrel is being kicked. Which one?’

(6) *Test sentence with verb* osu *‘push’*:

a.	Inusan-ga	osi-teru	yo.	Dotti?	(Active)
	dog-Nom	push-ing	excl	which	
	‘The dog is pushing *pro*. Which one?’				
b.	Inusan-ga	os-are-teru	yo.	Dotti?	(Passive)
	dog-Nom	push-Pass-ing	excl	which	
	‘The dog is being pushed. Which one?’				

By adding a question phrase *dotti?* ‘which one?’ at the end of each stimulus, we ensured that the child knew that she was expected to look at one or the other of the panels during the sessions. The order of the 16 trials was randomized per participant. The congruent events appeared on either side of the display in pseudo-random orders.

In addition, parents of the 2-3-year-olds were asked to fill out the Japanese MacArthur Communicative Development Inventory (Watamaki & Ogura 2004), which contains questions as to whether a child has already acquired certain words: The two target verbs are included in that list.

4.5. Results and analysis

We focused on the total fixation time on the two static images showing the outcome of the event, which appear during the last 5 seconds of each trial, starting with the second auditory stimulus and lasting until the end of the trial. By looking at the total fixation time data, we are able to infer how participants reach their interpretations after hearing the stimulus sentences twice. Previous off-line studies using the picture selection task, for example, can only provide us with information as to which picture each child has eventually chosen.

We excluded from further analysis the data of those who had not acquired the relevant verbs and children with few gaze samples or whose total fixation time on the two panels for at least one trial was zero, which left 32 2-3-year-olds (mean=2;11) and 19 6-year-olds (mean=6;4). We also had to exclude the data of one adult since she told the experimenter just after the experiment that she had consistently looked at the incongruent panels throughout the trials on purpose, questioning whether the task had some hidden tricks.

To analyze the data statistically, we first fit a linear mixed effects model with fixation time as the dependent variable and congruency, the active/passive distinction and age group as three fixed predictor variables. All the interaction terms between the three fixed factors, as well as a random intercept for participants, were included in the model. As we can see in Figures 2, 3, and 5, participants from different age groups appear to exhibit different patterns.

This observation manifests itself as a significant three-way interaction term (β=-0.74, *s.e.=0.30, t=-2.49, p<.05*). Since this complex model is difficult to interpret, we fit a simpler model to the data in each age group.[3]

Figure 2 shows the total fixation time by adults in the test session. In this graph, the vertical axis indicates total fixation times in the last 5 seconds. The gray boxplots indicate “Congruent,” which represents the duration of time when the participants were looking at the correct panels, while the white “Incongruent” boxplots show how long the participants were looking at the incorrect ones. We can see that the adults looked at the congruent scenes longer than the incongruent ones for both active and passive sentences. A linear mixed effects model with congruency and active/passive as the fixed factors, together with a random intercept for speakers, shows that congruency had a significant impact on total fixation time (β=3.59, *s.e.=0.14, t=26.3, p<.001*), whereas the active/passive distinction and its two-way interaction with congruency did not (β=0.07, *s.e.=0.14, t=0.54, n.s;* β=0.27, *s.e.=0.27, t=1.00, n.s.*).

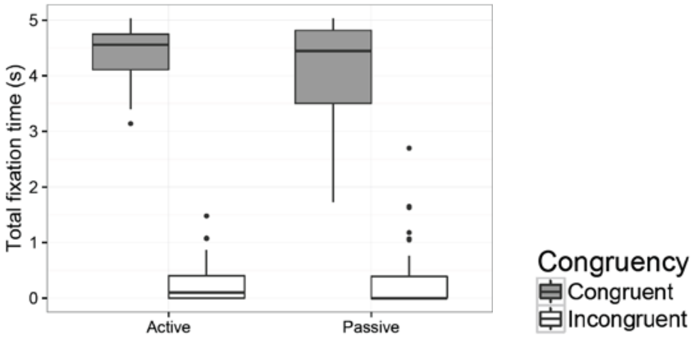

Figure 2: Total fixation time by adults[4]

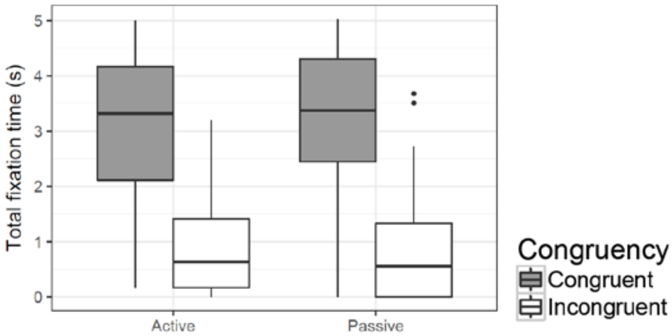

Figure 3: Total fixation time by 6-year-olds

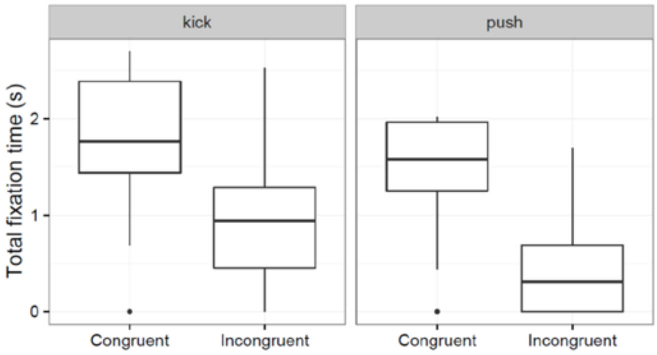

Figure 4: Total fixation time by 2-3-year-olds in pretest

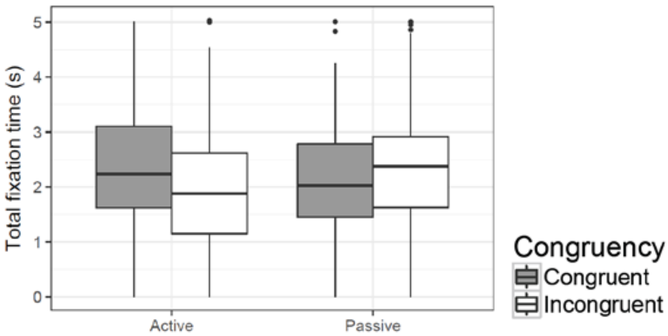

Figure 5: Total fixation time by 2-3-year-olds

The results for the 6-year-olds are given in Figure 3, where we see a pattern similar to the results for the adults, although the trend is less clear-cut. The 6-year-olds looked longer at the matching scenes for both active and passive sentences. For the 6-year-olds' data, a linear mixed effects model with congruency and active/passive as fixed factors revealed that the only significant factor was the congruent/incongruent distinction ($\beta=2.3$, $s.e.=0.12$, $t=19.48$, $p<.001$). Neither the active/passive distinction nor the interaction term was significant ($\beta=0.01$, $s.e.=0.12$, $t=0.07$, *n.s.*; $\beta=-0.12$, $s.e.=0.24$, $t=-0.49$, *n.s.*).

As for the children aged 2-3, let us consider first the results of the pretest section. Figure 4 shows that the children looked at the congruent image longer than the incongruent one to a statistically significant degree ($\beta=-0.81$, $s.e.=0.15$, $t=-5.57$, $p<.001$). The verb *kick* has a slight advantage over the verb *push* ($\beta=-0.30$, $s.e.=0.15$, $t=-2.04$, $p<.05$), but the interaction was non-significant ($\beta=-0.24$, $s.e.=0.21$, $t=-1.16$, *n.s.*). This means that they were able to connect the verbs *kick* and *push* with their matching scenes. In contrast, the total fixation times in the test sessions, as seen in Figure 5, indicate that passive sentences have slightly longer incongruent times, whereas active sentences have longer congruent times. A linear mixed model with congruency and active/passive as fixed factors with a random intercept of speakers revealed a significant interaction between congruency and the active/passive distinction (*β=0.62, s.e.=0.19, t=3.33, p<.001*). Neither congruency nor the active/passive distinction had a significant impact on total fixation time (*β=0.06, s.e.=0.09, t=0.69, n.s.; β=-0.08, s.e.=0.09, t=-0.87, n.s.*). Separate models exploring the effects of congruency in each of the active and passive conditions show that the 2-3-year-olds looked at congruent scenes longer than incongruent scenes for the active sentences (*β=0.38, s.e.=0.13, t=2.79, p<.01*), while no significant differences were identified in the passive condition (*β =-0.24, s.e.=0.13, t=-1.91, n.s.*). These results present a stark contrast from those of the 6-year-olds and the adults.

5. Discussion

The results of the current experiment revealed that the 2-3-year-old children looked longer at the congruent events for active sentences, whereas they tended to look at the incongruent ones longer for passive sentences. This asymmetry suggests that they comprehended active sentences but had difficulty with passive sentences. We also found that the 6-year-old children showed different behaviors from the 2-3-year-olds: they consistently looked longer at congruent scenes regardless of sentence types, performing almost as well as the adults did.

There are several possible causes for the non-adult-like behavior of the 2-3-year-old children. First, it is possible that the young children did not understand that they were supposed to look at the scene which they thought matched the stimulus sentences, even though we added the guiding phrase ‘which one?’ at the end of each stimulus. We can deny this possibility when we look at their performance during the pretest. We have already confirmed in the previous section (Figure 4) that they correctly looked at the congruent images much longer than the incongruent images in the pretest. This result suggests that their poor performance is not attributable to an inability to understand the task itself.

Another possible cause of the non-adult-like behavior of the 2-3-year-old children for passive sentences may be Lexical-ordering Strategy or the agent-first strategy (Bever 1970; Suzuki 1977). It has been suggested that this strategy is typically employed by young children who have trouble interpreting sentences with non-canonical word order, such as passive sentences or sentences with scrambling. Those children tend to regard the first NP as the actor. It is possible that the 2-3-year-old children in our experiment also used this strategy, which led them to interpret the first NPs of the passive sentences as actors rather than as actees. However, this explanation is not fully tenable when we compare their performances on passives and actives. If the children regarded the first NPs of passive sentences as actors, treating them just as they would in active sentences, the results for passives and actives would have been perfect mirror images. On the contrary, as was pointed out in the previous section (Figure 5), the distinction between congruent and incongruent total fixation times for passives is not statistically significant, but it is for actives. Thus, the agent-first strategy alone does not seem to tell the full story.

It is reasonable then to conjecture that the young children recognized the passive morpheme and thought that the passive sentences they had heard were somehow different from the active sentences, although they could not react promptly and confidently to the passive sentences within a few seconds. This account is along the same lines as the idea proposed by Omaki & Lidz (2015) based on many previous studies on child processing. Huang et al. (2013), for example, examined how Mandarin-speaking 5-year-olds interpret passive sentences with a referential subject, such as *Seal BEI it eat* 'The seal is eaten by it' (where *BEI* is a passive morpheme and is followed by an Agent NP and a VP (Huang 1999)), and with a pronominal subject, such as *It BEI seal eat* 'It is eaten by the seal.' The children interpreted both types of passive at around chance levels, but they comprehended sentences with a pronominal subject better than those with a referential subject. Huang et al. (2013) argue that the agent-first interpretation does not allow pronominal subjects of passives, and children's difficulties in comprehending passives were thus mitigated. Conversely, they had difficulty in revising the agent-first interpretation bias when the subjects of passives were expressed nouns. Omaki & Lidz (2015) argue that children's difficulties in comprehending certain constructions may be partly due to their immature sentence revision mechanisms. We can apply this assumption to the results of our experiment: The sentence revision mechanisms of the youngest participants in our experiment are not mature enough to succeed in revising the initial misinterpretations induced by the Lexical-ordering strategy even after they have realized that the sentence they heard had a passive morpheme. If they solely relied on the Lexical-ordering strategy, completely ignoring the passive morpheme and treating the sentences as if they were actives, they would have looked statistically longer at the incongruent scenes. However, the data showed otherwise. We thus conjecture that the child participants, whose sentence revision mechanisms are still not fully matured, could not decide confidently on which scene to watch and whether to cancel their first interpretation during the limited time window in the experimental condition. This is the most reasonable explanation for the observed behaviors of the 2-3-year-old children with passive sentences based on the results we have at hand.

6. Conclusion

The present study on comprehension of Japanese passives revealed that the participants of ages 2-3 looked longer at the congruent events when given active sentences, while they looked at the incongruent ones only slightly but not significantly longer when they heard passive sentences. We argued that their non-adult-like behavior for passive sentences cannot be accounted for only in terms of the difficulty of the experimental task or by the Lexical-ordering strategy. We claimed that although young children could not revise their initial interpretations guided by the agent-first strategy as quickly as 6-year-olds or adults, they were aware of the existence of passive morphemes and that those sentences they heard were different from the active ones. Our current account is compatible with the idea that young children have difficulty with sentence revision, as has been observed with various constructions.

Acknowledgements

This study was supported by JSPS KAKENHI Grant Number JP17K02711.

Notes

1 The abbreviations used in the glosses throughout this study are: NOM = nominative case, ACC = accusative case, Past = past tense, and Pass = passive suffix.

2 Since the attention span of 2-3-year-old children is generally short, we judged that 16 stimuli in total was the maximum for the young children.

3 An R markdown file with complete model details for all the analyses reported below is available upon request.

4 The thick black lines represent the medians. The upper edge of the box represents the 75th percentile, the lower edge the 25th percentile. The length of the whiskers is defined as 1.5 times interquartile range; dots were data points outside that range.

References

Armon-Lotem, Sharon, Ewa Haman, Kristine Jensen de López, Magdalena Smoczynska, Kazuko Yatsushiro, Marcin Szczerbinski, Angeliek van Hout, Ineta Dabašinskienė, Anna Gavarró, Erin Hobbs, Laura Kamandulytė-Merfeldienė, Napoleon Katsos, Sari Kunnari, Chrisa Nitsiou, Lone Sundahl Olsen, Xavier Parramon, Uli Sauerland, Reeli Torn-Leesik and Heather van der Lely (2016) A large-scale cross-linguistic investigation of the acquisition of passive, *Language Acquisition*, 23:1, 27-56.

Babyonyshev, Maria and Dina Brun (2004) The acquisition of perfective and imperfective passive constructions in Russian, *University of Pennsylvania Working Papers in Linguistics* 10, 17-31.

Bever, Thomas G. (1970) The cognitive basis for linguistic structures, *Cognition and Language Development*, ed. by John R. Hayes, 279-352, NY: Wiley.

Borer, Hagit and Kenneth Wexler (1987) The maturation of syntax, *Parameter Setting*, ed. by Thomas Roeper and Edwin Williams, 123-172, Reidel, Dordrecht.

Deen, Kamil Ud, Ivan Bondoc, Amber Camp, Sharon Estioca, Haerim Hwang, Gyu-Ho Shin, Maho Takahashi, Fred Zenker, and Jing Crystal Zhong (2018) Repetition brings success: Revealing knowledge of the passive voice, *Proceedings of the 42nd Annual Boston University Conference on Language Development*, ed. by Anne B. Bertolini and Maxwell J. Kaplan, 200-213, Cascadilla Press, Somerville, MA.

Demuth, Katherine, Francina Moloi and Malillo Machobane (2010) 3-year-olds' comprehension, production, and generalization of Sesotho passives, *Cognition* 115, 238-251.

Fox, Danny & Yosef Grodzinsky (1998) Children's passive: A view from the *by*-phrase, *Linguistic Inquiry* 29, 311-332.

Goodluck, Helen (1996) The act out task, *Methods for Assessing Children's Syntax*, ed. by Dana McDaniel, Cecil McKee and Helen Smith Cairns, 147-162, MIT Press, Cambridge, MA.

Hakuta, Kenji (1982) Interaction between particles and word order in the comprehension and production of simple sentences in Japanese children, *Developmental Psychology* 18, 62-76.

Harada, Kazuko & Tomoko Furuta (1999) On the maturation of A-chains: A view from Japanese passives, *COE Research Report* 3, Kanda University of International Studies, 397-426.

Huang, C.-T. James (1999) Chinese passives in comparative perspective, *Tsing Hua Journal of Chinese Studies* 29, 423-509.

Huang, Yi Ting, Xiaobei Zheng, Xiangzhi Meng and Jesse Snedeker (2013) Children’s assignment of grammatical roles in the online processing of Mandarin passive sentences, *Journal of Memory and Language* 69, 589-606.

Ishikawa, Megumi, Takuya Goro and Takane Ito (2018) Nihongo-zi-ni okeru tadosi-ukemibun-no rikai [Comprehension of transitive passives in Japanese children], Paper presented at the 156th Meeting of the Linguistic Society of Japan.

Jaeggli, Osvaldo (1986) Passive, *Linguistic Inquiry* 17, 587-622.

Kubo, Miori (1992) Japanese passives. *Gengo Bunka Kiyoo* 23, 231-302, Hokkaido University.

Minai, Utako (2000) The acquisition of Japanese passives, *Japanese/Korean Linguistics* 9, 339-350.

Murasugi, Keiko and Tomoko Kawamura (2005) On the acquisition of scrambling in Japanese, *The Free Word Order Phenomenon: Its Syntactic Sources and Diversity*, ed. by Joachim Sabel and Mamoru Saito, 335-376, Mouton de Gruyter, Berlin.

O’Brien, Karen, Elaine Grolla and Diane Lillo-Martin (2006) Long passives are understood by young children, *Proceedings of the 30th Annual Boston Conference on Language Development*, ed. by David Bamman, Tatiana Magnitskaia, and Colleen Zaller, 441-451, Cascadilla Press, Somerville, MA.

Okabe, Reiko and Tetsuya Sano (2002) The acquisition of implicit arguments in Japanese and related matters, *Proceedings of the 26th Annual Boston Conference on Language Development*, ed. by Barbora Skarabela, Sarah Fish, and Anna H.-J. Do, 485-499, Cascadilla Press, Somerville, MA.

Omaki, Akira and Jeffrey Lidz (2015) Linking parser development to acquisition of syntactic knowledge, *Language Acquisition* 22, 158-192.

Pierce, Amy (1992) The acquisition of passives in Spanish and the question of A-chain maturation, *Language Acquisition* 2, 55-81.

Sano, Keiko (1977) An experimental study on the acquisition of Japanese simple sentences and cleft sentences, *Descriptive and Applied Linguistics* 10, 213-233, International Christian University, Tokyo.

Sano, Tetsuya (2013) Remarks on theoretical accounts of Japanese children's passive acquisition, *Generative Linguistics and Acquisition: Studies in Honor of Nina M. Hyams*, ed. by Misha Becker, John Grinstead and Jason Rothman, 35-64, Amsterdam: John Benjamins.

Sano, Tetsuya, Mika Endo and Kyoko Yamakoshi (2001) Developmental issues in the acquisition of Japanese unaccusatives and passives, *Proceedings of the 25th Annual Boston Conference on Language Development*, ed. by Anna H.-J. Do, Laura Domínguez, and Aimee Johansen, 668-683, Cascadilla Press, Somerville, MA.

Shibatani, Masayoshi (1978) *Nihongo no Bunseki* [Analysis of Japanese], Tokyo: Taishukan.

Sugisaki, Koji (1999) Japanese passives in acquisition, *UConn Working Papers in Linguistics* 10, 145-156.

Suzuki, Seiichi (1977) Nihon no yozi ni okeru gozyun horyaku [Lexical-ordering strategy in early Japanese children], *The Japanese Journal of Educational Psychology* 25:3, 56-61.

Terzi, Arhonto and Ken Wexler (2002) A-chains and s-homophones in children's grammar: Evidence from Greek passives. *Proceedings of NELS* 32, 519-537.

Watamaki, Toru and Tamiko Ogura (2004) *Technical Manual of the Japanese MacArthur Communicative Development Inventory: Words and Grammar*, Kyoto International Social Welfare Exchange Center.